

Supplementary Material: Policy-Specific Convergence and Regret Results

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Appendix A: Game, Convergence, and Regret

A.1. Scope

This document provides the results for three exact self-play processes in the unit-translated Smead finite- N quasi-centipede, normalized to the notebook matrix: repaired anytime Exp3, zero-initialized sample-mean epsilon-greedy, and Gaussian Thompson Sampling. I prove unconditional convergence for repaired anytime Exp3 under its stated schedule. Epsilon-greedy and Thompson Sampling require a calibration condition because they estimate action values from different histories of the opponent's play. Two-stage countertheorems rule out the old unconditional expected logarithmic-regret claims for both policies.

Both players use the same policy. Nothing below covers pairings of different policies. The probability statements are supplied-process results. I assume a filtered probability space carrying the stated policy, adapted actions, and conditionally independent private randomization by the two players at each round.

I keep three conclusions separate. Selection-probability convergence means that the probability of selecting stage 0 tends to one. Empirical convergence means that the frequency of $(0,0)$ tends to one. Neither conclusion makes the realized action sequence constant. Under the policies and schedule ranges below, every stage is selected infinitely often even when the probabilities of positive stages tend to zero.

A.2. The finite quasi-centipede game

Let $N \geq 2$ and $S = \{0, \dots, N-1\}$. Each player chooses an intended take-stage. From either player's perspective, choosing stage k against stage j gives the raw payoff

$$u(k, j) = \begin{cases} 2k + 5, & k < j, \\ 2k + 2, & k = j, \\ 2j + 1, & k > j. \end{cases} \quad (\text{A.1})$$

For $N = 3$, the payoff matrix for either player is

$$M_3 = \begin{pmatrix} 2 & 5 & 5 \\ 1 & 4 & 7 \\ 1 & 3 & 6 \end{pmatrix}. \quad (\text{A.2})$$

Smead's (2008) printed three-stage matrix is

$$M_3^{\text{Smead}} = \begin{pmatrix} 1 & 4 & 4 \\ 0 & 3 & 6 \\ 0 & 2 & 5 \end{pmatrix}. \quad (\text{A.3})$$

The manuscript and notebook add one to every payoff, producing (A.1)-(A.2): $u^{\text{Smead}}(k, j) = u(k, j) - 1$. The reason is historical. The 2024 implementation used Beta-Bernoulli Thompson Sampling and converted each reward into 1 when it exceeded 1 and 0 otherwise. Gaussian Thompson Sampling no longer needs that encoding, but the empirical results and formal theorems use the notebook matrix. I retain it here.

The translation preserves payoff differences, dominance relations, equilibria, and raw external-regret differences. It disappears from repaired Exp3 because the normalization $g = (u - 1)/(2N)$ in (B.1) returns the payoffs to Smead's printed scale. Epsilon-greedy and Gaussian Thompson Sampling are different: their initial estimates and prior means are absolute values. Translating the payoffs preserves their finite paths only if those initial values move by the same amount. The notebook keeps its original initialization, so the theorems for those policies concern the unit-translated game.

For an opponent distribution $p \in \Delta(S)$, write

$$U(k, p) = \sum_{j=0}^{N-1} p(j)u(k, j). \quad (\text{A.4})$$

Direct subtraction gives the adjacent payoff identity

$$U(k - 1, p) - U(k, p) = p(k - 1) + p(k) - 2 \sum_{j=k+1}^{N-1} p(j). \quad (\text{A.5})$$

The final term in (A.5) prevents a global adjacent-dominance claim. Stage $k - 1$ need not outperform stage k while the opponent still places enough probability on higher stages. Once those stages have been removed, the final term vanishes. In the reduced game $\{0, \dots, k\}$, stage $k - 1$ weakly dominates stage k and does strictly better whenever the opponent places positive probability on stage $k - 1$ or k .

Proposition A.1 (Game-level solution). *The finite quasi-centipede game is solved by descending iterated elimination of weakly dominated stages and has the unique Nash equilibrium $(0, 0)$.*

Proof. In the reduced game $\{0, \dots, k\}$, stage $k - 1$ ties stage k against every column below $k - 1$ and beats it by one against columns $k - 1$ and k . Eliminating $N - 1, N - 2, \dots, 1$ therefore leaves stage 0.

For uniqueness, let r_i be the largest stage in Player i 's equilibrium support. If the opponent's largest support stage is $r > 0$, stage $r - 1$ strictly beats stage r because the opponent assigns positive probability to column r . Stage r also strictly beats every stage above it against that opponent distribution. Every best response must therefore have a largest support stage below r . Equilibrium would require both $r_1 < r_2$ and $r_2 < r_1$, which is impossible. If either largest support stage is 0, the other player's unique best response is 0. $\square$

At each step, the learning proofs need one comparison that remains fixed while the values of lower stages continue to change. The mixture below provides it.

Lemma A.2 (Fixed reduced-game dominators). *For every $k \in \{1, \dots, N - 1\}$, there is a distribution $\lambda^{(k)}$ supported on $\{0, \dots, k - 1\}$ such that*

$$\sum_{a < k} \lambda^{(k)}(a)u(a, j) > u(k, j) \quad (0 \leq j \leq k). \quad (\text{A.6})$$

Proof. Put $\lambda^{(1)} = e_0$. Stage 0 beats stage 1 by one against both reduced-game columns. Suppose $\lambda^{(k-1)}$ has been constructed. Define

$$V_k = \sum_{a < k-1} \lambda^{(k-1)}(a)(2a + 5), \quad \epsilon_k = \frac{1}{2(2k + 3 - V_k)}, \quad (\text{A.7})$$

and

$$\lambda^{(k)} = (1 - \epsilon_k)e_{k-1} + \epsilon_k\lambda^{(k-1)}. \quad (\text{A.8})$$

The support of $\lambda^{(k-1)}$ lies below $k - 1$, so $V_k \leq 2k + 1$ and $0 < \epsilon_k \leq 1/4$. Against a column below $k - 1$, stages $k - 1$ and k tie, while the inherited mixture supplies a strict improvement. Against column $k - 1$, stage $k - 1$ already beats stage k . Against column k , the new mixture's

advantage is

$$1 - \epsilon_k(2k + 3 - V_k) = \frac{1}{2}. \quad (\text{A.9})$$

The mixture therefore beats stage k on every column of the reduced game. $\square$

For later use, define the strictly positive reduced-game gap

$$\delta_k = \min_{0 \leq j \leq k} \left[\sum_{a < k} \lambda^{(k)}(a) u(a, j) - u(k, j) \right] > 0. \quad (\text{A.10})$$

A.3. Convergence objects and the regret comparator

Rounds are indexed by $t = 0, 1, \dots$. Let $A_{i,t}$ be Player i 's realized stage and let $\pi_{i,t}(a) = \Pr(A_{i,t} = a \mid \mathcal{F}_t)$ be the selection probability before round t . Stage-zero selection convergence means

$$\pi_{i,t}(0) \longrightarrow 1. \quad (\text{A.11})$$

Player i 's empirical distribution and the empirical joint distribution over the half-open horizon $[0, T)$ are

$$F_{i,T}(a) = \frac{1}{T} \sum_{t < T} \mathbf{1}\{A_{i,t} = a\}, \quad (\text{A.12})$$

$$F_T(a, b) = \frac{1}{T} \sum_{t < T} \mathbf{1}\{A_{1,t} = a, A_{2,t} = b\}. \quad (\text{A.13})$$

The regret results use raw external regret against the best fixed pure stage on the opponent's realized action sequence:

$$R_i^{\text{ext}}(T) = \max_{a \in S} \sum_{t < T} [u(a, A_{-i,t}) - u(A_{i,t}, A_{-i,t})]. \quad (\text{A.14})$$

Equation (A.14) asks which single stage would have paid most against the opponent actions that actually occurred. Its maximum therefore remains inside every pathwise expression and any later expectation. The comparison does not change the opponent's realized sequence, so it is not policy regret. Nor is it equilibrium payoff shortfall. A run can outperform repeated $(0, 0)$ and still lose value relative to another fixed stage.

A.4. Why weak dominance does not settle the learning problem

Iterated elimination holds the opponent distribution fixed while it compares two stages. Sample-mean epsilon-greedy and Gaussian Thompson Sampling do not. Each stage is updated only when selected, so its estimate averages the opponent behavior observed on that stage's own action clock.

For Player i , define the count and joint count

$$N_{i,a}(T) = \sum_{t < T} \mathbf{1}\{A_{i,t} = a\}, \quad J_{i,a,b}(T) = \sum_{t < T} \mathbf{1}\{A_{i,t} = a, A_{-i,t} = b\}. \quad (\text{A.15})$$

Whenever $N_{i,a}(T) > 0$, the opponent distribution observed while Player i selected a is

$$\nu_{i,a,T}(b) = \frac{J_{i,a,b}(T)}{N_{i,a}(T)}. \quad (\text{A.16})$$

Different actions can therefore be evaluated against different $\nu_{i,a,T}$. Infinite sampling makes every denominator diverge, but it does not make the distributions in (A.16) agree. In the two-stage game, the sample-mean contrast has the exact form

$$Q_{i,T}(0) - Q_{i,T}(1) = 1 + 3[\nu_{i,0,T}(1) - \nu_{i,1,T}(1)]. \quad (\text{A.17})$$

Stage 0 beats stage 1 by one against every common opponent distribution. Equation (A.17) isolates the problem: the two estimates need not use a common distribution. Appendices C and D state the calibration needed to align them. Exp3 avoids this mismatch because its importance-weighted estimates produce conditionally unbiased mixed payoff contrasts at each round, before separate action clocks can form.

Appendix B: Repaired Anytime Exp3

B.1. Policy and schedules

Exp3 uses normalized gains

$$g(k, j) = \frac{u(k, j) - 1}{2N} \in [0, 1]. \quad (\text{B.1})$$

Let $w_{i,t}(a) > 0$ be Player i 's weight on stage a . Define the exploitation distribution and the actual selection distribution by

$$q_{i,t}(a) = \frac{w_{i,t}(a)}{\sum_r w_{i,t}(r)}, \quad \pi_{i,t}(a) = (1 - \gamma_t)q_{i,t}(a) + \frac{\gamma_t}{N}. \quad (\text{B.2})$$

After selecting $A_{i,t}$, Player i forms the importance-weighted estimate

$$\hat{g}_{i,t}(a) = \frac{\mathbf{1}\{A_{i,t} = a\}g(a, A_{-i,t})}{\pi_{i,t}(a)} \quad (\text{B.3})$$

and updates

$$w_{i,t+1}(a) = w_{i,t}(a) \exp\{\eta_t \hat{g}_{i,t}(a)\}. \quad (\text{B.4})$$

The repaired anytime schedules are

$$\gamma_t = d((t+1) + t_0)^{-b}, \quad \eta_t = c((t+1) + t_0)^{-2b}, \quad (\text{B.5})$$

where $c, d > 0$, $t_0 \geq 0$, and $0 < \gamma_t < 1$ at every round.

B.2. Unconditional convergence

Theorem B.1 (Exp3 selection convergence). *Suppose both players use the policy in (B.2)-(B.5) in the unit-translated Smead finite- N quasi-centipede, normalized to the notebook matrix, with arbitrary strictly positive initial weights. If $0 < b \leq 1/2$, then, for both players,*

$$q_{i,t}(0) \longrightarrow 1, \quad \pi_{i,t}(0) \longrightarrow 1 \quad \text{almost surely.} \quad (\text{B.6})$$

Consequently, $F_{i,T}(0) \rightarrow 1$ and $F_T(0,0) \rightarrow 1$ almost surely.

Proof. Fix a positive stage k and use the mixture $\lambda^{(k)}$ from Lemma A.2. Define the importance-weighted mixed contrast

$$X_{i,k,t} = \sum_{a < k} \lambda^{(k)}(a) \hat{g}_{i,t}(a) - \hat{g}_{i,t}(k). \quad (\text{B.7})$$

Let $\mu_{i,k,t} = \mathbb{E}[X_{i,k,t} \mid \mathcal{F}_t]$ and $D_{i,k,t} = X_{i,k,t} - \mu_{i,k,t}$. Conditional independence of the players' current private draws gives

$$\mathbb{E}[\hat{g}_{i,t}(a) \mid \mathcal{F}_t] = \sum_j \pi_{-i,t}(j) g(a, j). \quad (\text{B.8})$$

Only one own action is selected at a round. The importance-weighted terms in (B.7) therefore have disjoint supports, and

$$\mathbb{E}[D_{i,k,t}^2 \mid \mathcal{F}_t] \leq \frac{2N}{\gamma_t}. \quad (\text{B.9})$$

Put $H_T = \sum_{t < T} \eta_t$ and define the log-geometric weight contrast

$$L_{i,k,T} = \sum_{a < k} \lambda^{(k)}(a) \log w_{i,T}(a) - \log w_{i,T}(k). \quad (\text{B.10})$$

The multiplicative update gives the exact recursion

$$L_{i,k,T} = L_{i,k,0} + \sum_{t < T} \eta_t \mu_{i,k,t} + \sum_{t < T} \eta_t D_{i,k,t}. \quad (\text{B.11})$$

The schedule satisfies $H_T \rightarrow \infty$ and

$$\sum_t \frac{\eta_t^2}{\gamma_t H_{t+1}^2} < \infty \quad (0 < b \leq 1/2). \quad (\text{B.12})$$

The square-integrable martingale theorem and Kronecker's lemma then give

$$\frac{\sum_{t < T} \eta_t D_{i,k,t}}{H_T} \longrightarrow 0 \quad \text{almost surely.} \quad (\text{B.13})$$

Let $\rho_{-i,k,t} = \sum_{j > k} \pi_{-i,t}(j)$. The mixed dominator has gain at least $\delta_k/(2N)$ on the reduced columns and gain no smaller than -1 elsewhere. Hence its conditional normalized-gain drift satisfies

$$\mu_{i,k,t} \geq \frac{\delta_k}{2N} (1 - \rho_{-i,k,t}) - \rho_{-i,k,t}. \quad (\text{B.14})$$

At the highest stage, $k = N - 1$, the upper tail is empty. Equation (B.14) then supplies a fixed positive drift, and (B.11)-(B.13) imply $L_{i,N-1,T} \rightarrow \infty$. Weighted AM-GM gives

$$q_{i,T}(k) \leq \frac{1}{1 + \exp(L_{i,k,T})}, \quad (\text{B.15})$$

so stage $N - 1$ vanishes for both players. Now suppose both players have eliminated every stage above k . We then have $\rho_{-i,k,t} \rightarrow 0$, and the drift in (B.14) becomes positive after a finite random time. The same argument removes stage k . Descending through the finitely many stages forces $q_{i,t}(0) \rightarrow 1$. The exploration rate also satisfies $\gamma_t \rightarrow 0$, so (B.2) gives the selection-probability conclusion.

Bounded martingale laws transfer selection-probability convergence to the marginal and joint empirical frequencies. $\square$

Theorem B.1 concerns probabilities and frequencies, not eventual constancy of the realized actions. The policy's exploration floor vanishes, but $\sum_t \gamma_t = \infty$ for $0 < b \leq 1/2$. Conditional Borel-Cantelli therefore gives infinitely many exploratory selections almost surely.

The endpoint cases explain why Theorem B.1 stops at $b = 1/2$. At $b = 0$, the uniform

exploration floor never vanishes, so $\pi_{i,t}(0)$ cannot tend to one. For the valid $t_0 = 0$ subfamily with $b > 1/2$, the learning updates are summable and every exploitation coordinate remains bounded away from zero almost surely. Thus no larger universal range is available within this power family.

B.3. Pathwise and expected external regret

Define the half-open exploration clock

$$\Gamma_T = \sum_{t < T} \gamma_t. \quad (\text{B.16})$$

For $0 < b < 1$, the shifted power-sum asymptotic is

$$\frac{\Gamma_T}{T^{1-b}} \longrightarrow \frac{d}{1-b}. \quad (\text{B.17})$$

Theorem B.2 (Exp3 pathwise external regret). *Under the assumptions of Theorem B.1, let $0 < b < 1/2$. For arbitrary strictly positive initial weights and each player,*

$$\frac{R_i^{\text{ext}}(T)}{\Gamma_T} \longrightarrow \frac{N-1}{N} \quad \text{almost surely.} \quad (\text{B.18})$$

Proof. In the interior range, the log contrast separates each positive stage from its lower-stage mixture at a rate set by H_T . After the finite onset at each level, the positive drift grows on that clock while the martingale term is $o(H_T)$. Every positive exploitation coordinate consequently has an eventual path-dependent stretched-exponential upper bound and

$$q_{i,t}(k) = o(\gamma_t) \quad (k > 0). \quad (\text{B.19})$$

Combining this rate with (B.2) gives

$$\pi_{i,t}(k) = \frac{\gamma_t}{N} + o(\gamma_t) \quad (k > 0). \quad (\text{B.20})$$

Predictable-count martingale laws yield

$$\frac{\sum_{t < T} \mathbf{1}\{A_{i,t} > 0\}}{\Gamma_T} \longrightarrow \frac{N-1}{N}, \quad (\text{B.21})$$

while current-round conditional independence gives

$$\frac{\sum_{t < T} \mathbf{1}\{A_{i,t} > 0, A_{-i,t} > 0\}}{\Gamma_T} \longrightarrow 0. \quad (\text{B.22})$$

The opponent chooses a nonzero stage only $o(T)$ times, so stage 0 is eventually the unique best fixed stage in hindsight. Against opponent stage 0, every own nonzero stage loses exactly one raw payoff unit to stage 0. The only discrepancy between that unit count and external regret occurs on simultaneous-nonzero rounds, which are negligible by (B.22). Equations (B.21)-(B.22) prove (B.18). $\square$

Theorem B.3 (Exp3 expected external regret). *Add $t_0 > 0$, literal equal unit initialization, $w_{i,0}(a) = 1$ for every player and stage, and the standard measurability assumptions of a Standard Borel probability space. If $0 < b < 1/2$, then*

$$\frac{\mathbb{E}[R_i^{\text{ext}}(T)]}{\Gamma_T} \longrightarrow \frac{N-1}{N}. \quad (\text{B.23})$$

Proof. Equal unit initialization fixes every initial log contrast at zero. One-sided martingale concentration applied to (B.11), followed by the same finite downward induction as in Theorem B.1, gives a summable expected bound on every positive exploitation coordinate:

$$\sum_{t=0}^{\infty} \mathbb{E}[q_{i,t}(k)] < \infty \quad (k > 0). \quad (\text{B.24})$$

The own nonzero count then has expectation $((N-1)/N)\Gamma_T + O(1)$, while simultaneous nonzero play contributes only $o(\Gamma_T)$ in expectation. What remains is the hindsight maximum in (A.14). Write raw external regret as regret to fixed stage 0 plus the nonnegative correction for whichever fixed stage performs best on the realized opponent sequence. The opponent's expected positive exploitation mass is summable. Its nonzero-action count has a predictable compensator with an $O(\Gamma_T) = o(T)$ uniform-exploration term, and the centered count is a bounded martingale. Markov and Azuma bounds make a linear number of nonzero opponent actions rare enough that the expected hindsight correction stays uniformly bounded. Division by $\Gamma_T \rightarrow \infty$ proves (B.23). $\square$

Arbitrary positive random initial weights are sufficient for Theorems B.1 and B.2, but not for Theorem B.3. Heavy-tailed initial log-weight contrasts can preserve pathwise convergence while making expected regret grow faster than the exploration clock. The equal-unit assumption in Theorem B.3 therefore cannot be omitted.

At $b = 1/3$, the exploration clock in (B.17) grows as $3dT^{2/3}/2$. Theorems B.2 and B.3 therefore give

$$\frac{R_i^{\text{ext}}(T)}{T^{2/3}} \longrightarrow \frac{3d(N-1)}{2N} \quad \text{almost surely,} \quad (\text{B.25})$$

and, under equal unit initialization,

$$\frac{\mathbb{E}[R_i^{\text{ext}}(T)]}{T^{2/3}} \longrightarrow \frac{3d(N-1)}{2N}. \quad (\text{B.26})$$

These limits belong to the repaired anytime policy. Standard horizon-fixed Exp3 with a known terminal horizon is a different policy. Auer, Cesa-Bianchi, Freund, and Schapire (2002) prove a square-root external-regret bound for fixed or exogenous bounded reward sequences under their stated comparator and expectation order. In adaptive self-play, their argument controls each fixed comparator in expectation, but it does not by itself move the realized hindsight maximum inside the expectation. Neither statement transfers to the repaired schedule in (B.5) merely by changing the schedule's name.

Appendix C: Epsilon-Greedy

C.1. Exact process and action-clock regularity

For each player and stage, let $Q_{i,t}(a)$ be the lifetime sample mean before round t , initialized at zero. Let

$$G_{i,t} = \arg \max_{a \in S} Q_{i,t}(a) \quad (\text{C.1})$$

be the nonempty greedy set, with uniform tie-breaking. Put $\varepsilon_t = 1/(t+1)$. The selection probabilities are

$$\pi_{i,t}(a) = (1 - \varepsilon_t) \frac{\mathbf{1}\{a \in G_{i,t}\}}{|G_{i,t}|} + \frac{\varepsilon_t}{N}. \quad (\text{C.2})$$

Only the selected stage is updated by its realized payoff. At $t = 0$, the policy selects uniformly.

The floor $1/[N(t+1)]$ makes every action's conditional-probability sum diverge. Conditional Borel-Cantelli therefore gives infinite sampling almost surely. More is needed. Define the predictable own-action clock

$$P_{i,a}(T) = \sum_{t < T} \pi_{i,t}(a). \quad (\text{C.3})$$

On one probability-one event, for every player and action pair,

$$\frac{N_{i,a}(T)}{P_{i,a}(T)} \longrightarrow 1, \quad (\text{C.4})$$

and

$$\frac{J_{i,a,b}(T) - \sum_{t < T} \pi_{i,t}(a)\pi_{-i,t}(b)}{P_{i,a}(T)} \longrightarrow 0. \quad (\text{C.5})$$

For (C.4), subtract the predictable count $P_{i,a}(T)$ from the realized count $N_{i,a}(T)$. The difference is a bounded martingale whose conditional variance is controlled by $P_{i,a}(T)$. The corresponding calculation for $J_{i,a,b}(T)$ uses $\pi_{i,t}(a)\pi_{-i,t}(b)$, the product probability supplied by the players' conditionally independent current draws. Self-normalized martingale laws give (C.4)-(C.5). We now know that every stage is sampled infinitely often and that the own and joint action-clock laws hold almost surely. We do not yet know that different actions observe the same history of opponent play.

C.2. The calibration condition

For the mixed dominator $\lambda^{(k)}$, define the estimated mixed contrast

$$C_{i,k}(T) = \sum_{a < k} \lambda^{(k)}(a) Q_{i,T}(a) - Q_{i,T}(k), \quad (\text{C.6})$$

and the corresponding contrast against the opponent's global empirical distribution

$$D_{i,k}(T) = \sum_{a < k} \lambda^{(k)}(a) U(a, F_{-i,T}) - U(k, F_{-i,T}). \quad (\text{C.7})$$

Let $\mathcal{D}_{\text{mix}}$ be the event on which, for both players and every positive stage,

$$\liminf_{T \rightarrow \infty} [C_{i,k}(T) - D_{i,k}(T)] \geq 0. \quad (\text{C.8})$$

I call $\mathcal{D}_{\text{mix}}$ the mixed-clock calibration event. It is one-sided. The condition does not require every action estimate to converge to a common-clock value; it requires only that differences among the action clocks do not erase the mixed-dominator advantage used in the next step of the downward induction. Whether this event occurs remains unproved.

C.3. Conditional convergence

Theorem C.1 (Epsilon-greedy conditional convergence). *For every finite $N \geq 2$ and every supplied exact zero-initialized same-policy epsilon-greedy process in the unit-translated Smead finite- N quasi-centipede, normalized to the notebook matrix, almost every path has the following property: if $\mathcal{D}_{\text{mix}}$ holds, then there is a finite time after which both greedy sets equal $\{0\}$. Consequently,*

$$\pi_{i,t}(0) = 1 - \frac{N-1}{N(t+1)}, \quad \pi_{i,t}(a) = \frac{1}{N(t+1)} \quad (a > 0) \quad (\text{C.9})$$

for all sufficiently large t , and $\pi_{i,t}(0) \rightarrow 1$, $F_{i,T}(0) \rightarrow 1$, and $F_T(0,0) \rightarrow 1$.

Proof. Work on the probability-one event carrying (C.4)-(C.5), and suppose $\mathcal{D}_{\text{mix}}$ holds. Begin with $k = N-1$, where every opponent stage lies in the reduced game. Lemma A.2 gives $D_{i,N-1}(T) \geq \delta_{N-1}$ at every horizon, and (C.8) makes the estimated mixed contrast eventually positive. A convex combination of lower-stage estimates then exceeds $Q_{i,T}(N-1)$. At least one lower stage has a strictly larger estimate, so stage $N-1$ leaves the greedy set permanently.

Now suppose stages above k have left both greedy sets. They are selected only through uniform exploration, with probability $1/[N(t+1)]$, and (C.4)-(C.5) make their empirical frequencies vanish. Their negative contribution to (C.7) vanishes with them. Lemma A.2 leaves a reduced-game advantage of at least δ_k , which calibration transfers to (C.6). Stage k then leaves both greedy sets. Repeating the argument removes every positive stage. Equation (C.9) and the marginal and joint martingale laws give the remaining conclusions. $\square$

Nothing in the proof establishes that $\mathcal{D}_{\text{mix}}$ occurs. Infinite sampling and the action-clock laws supply the regularity used by the implication, but not its antecedent. Unconditional epsilon-greedy convergence therefore remains open for this exact process.

C.4. Regret after greedy-set settlement

Let $\mathcal{G}$ be the event that both greedy sets permanently equal $\{0\}$ after some finite time.

Theorem C.2 (Epsilon-greedy settled-branch regret). *For each player, almost every path satisfies*

$$\mathcal{G} \implies \frac{R_i^{\text{ext}}(T)}{\log T} \longrightarrow \frac{N-1}{N}. \quad (\text{C.10})$$

Proof. A future settlement event cannot be inserted directly into a one-step conditional expectation. Stop the profile counts at the first later exit from $\{0\}$, derive their predictable compensators from (C.2), and apply bounded stopped-martingale laws. On every settled path, each profile $(a,0)$ with $a > 0$ occurs

$$\frac{1}{N} \log T + o(\log T) \quad (\text{C.11})$$

times, while simultaneous nonzero play occurs only finitely often. The opponent's nonzero frequency is $O(\log T/T)$, so stage 0 is eventually the unique best fixed stage in hindsight. Each own-only deviation $(a,0)$ loses exactly one raw payoff unit to stage 0. Summing (C.11) over

the $N - 1$ positive stages and discarding the finite simultaneous-nonzero contribution proves (C.10). $\square$

The open question is whether $\mathcal{G}$ occurs. Theorem C.2 is a pathwise implication and supplies no unconditional expected logarithmic bound.

C.5. The two-stage expected-regret countertheorem

At $N = 2$, stage 0 beats stage 1 by one against either opponent stage. Thus

$$R_i^{\text{ext}}(T) = N_{i,1}(T) \quad (\text{C.12})$$

on every path. Let $E_{i,T}$ be the event that Player i selects stage 1 in every round $t < T$. After the first positive payoff on stage 1, that stage remains uniquely greedy while the unplayed stage 0 retains value zero. Therefore

$$\Pr(E_{i,T}) = \prod_{t < T} \frac{2t+1}{2t+2} \geq \frac{1}{2\sqrt{T}} \quad (T > 0). \quad (\text{C.13})$$

On $E_{i,T}$, external regret equals T .

Theorem C.3 (Epsilon-greedy expected-regret lower bound). *In the exact two-stage zero-initialized same-policy process, for either player and every positive horizon,*

$$\mathbb{E}[R_i^{\text{ext}}(T)] \geq \frac{\sqrt{T}}{2}. \quad (\text{C.14})$$

Consequently, there are no constants $C \geq 0$ and T_0 for which $\mathbb{E}[R_i^{\text{ext}}(T)] \leq C \log T$ for every $T \geq T_0$.

Proof. Equations (C.12)-(C.13) give

$$\mathbb{E}[R_i^{\text{ext}}(T)] \geq T \Pr(E_{i,T}) \geq \frac{\sqrt{T}}{2}. \quad (\text{C.15})$$

The ratio $\sqrt{T}/\log T$ diverges, so an eventual logarithmic upper bound is impossible. $\square$

This lower bound leaves convergence open. Its point is that rare, long-lived paths prevent the conditional logarithmic law in Theorem C.2 from becoming an unconditional expected theorem.

Appendix D: Gaussian Thompson Sampling

D.1. Exact process and regularity

Fix a finite prior mean μ_0 , prior variance $\tau_0^2 > 0$, and pseudo-likelihood variance $\sigma^2 > 0$. Let

$$\rho = \frac{\sigma^2}{\tau_0^2}. \quad (\text{D.1})$$

For each player and stage, let $S_{i,a}(T)$ be the sum of the payoffs observed on that stage. The Gaussian pseudo-posterior before the next round has mean and variance

$$m_{i,a}(T) = \frac{S_{i,a}(T) + \rho\mu_0}{N_{i,a}(T) + \rho}, \quad v_{i,a}(T) = \frac{\sigma^2}{N_{i,a}(T) + \rho}. \quad (\text{D.2})$$

At each round, each player draws one independent Gaussian sample from every current pseudo-posterior and selects the stage with the largest draw. Gaussian ties have probability zero.

The notebook uses $\mu_0 = 0$, $\tau_0^2 = 100$, and the population variance of the entries of the N -stage payoff matrix,

$$\sigma_N^2 = \frac{2N^4 + 29N^2 - 27N - 4}{9N^2}.$$

The convergence and pure-branch results allow any finite prior mean and positive finite prior and pseudo-likelihood variances. The two-stage countertheorem returns to the notebook's literal values.

Every stage is sampled infinitely often almost surely. Suppose otherwise. After its final selection, a stage would retain a fixed nondegenerate Gaussian pseudo-posterior while all means and variances remained bounded. It would then keep a uniformly positive conditional probability of defeating the other Gaussian draws, contradicting its permanent avoidance. Equation (D.2) now implies that every posterior variance tends to zero.

Conditional current-round independence also gives the action-clock law

$$v_{i,a,T}(b) - \frac{\sum_{t < T} \pi_{i,t}(a) \pi_{-i,t}(b)}{\sum_{t < T} \pi_{i,t}(a)} \longrightarrow 0 \quad \text{almost surely.} \quad (\text{D.3})$$

As in Appendix C, (D.3) identifies each action's opponent distribution; it does not make the distributions equal across actions.

D.2. Conditional convergence

Define the posterior mixed contrast

$$C_{i,k}^{\text{TS}}(T) = \sum_{a < k} \lambda^{(k)}(a) m_{i,a}(T) - m_{i,k}(T), \quad (\text{D.4})$$

and retain the global empirical contrast $D_{i,k}(T)$ from (C.7). Let $\mathcal{D}_{\text{mix}}^{\text{TS}}$ be the event on which, for both players and every positive stage,

$$\liminf_{T \rightarrow \infty} [C_{i,k}^{\text{TS}}(T) - D_{i,k}(T)] \geq 0. \quad (\text{D.5})$$

The event $\mathcal{D}_{\text{mix}}^{\text{TS}}$ is the Gaussian Thompson counterpart of the epsilon-greedy calibration event.

Theorem D.1 (Thompson conditional convergence). *For every finite $N \geq 2$ and every supplied exact same-policy Gaussian Thompson process in the unit-translated Smead finite- N quasi-centipede, normalized to the notebook matrix, almost every path has the following property: if $\mathcal{D}_{\text{mix}}^{\text{TS}}$ holds, then*

$$\pi_{i,t}(0) \longrightarrow 1, \quad F_{i,T}(0) \longrightarrow 1, \quad F_T(0,0) \longrightarrow 1. \quad (\text{D.6})$$

On the same paths,

$$m_{i,0}(T) \longrightarrow 2, \quad m_{i,a}(T) \longrightarrow 1 \quad (a > 0). \quad (\text{D.7})$$

Proof. Begin with stage $N - 1$. Lemma A.2 and (D.5) give an eventual positive posterior mixed contrast, so some lower-stage posterior mean exceeds the mean of stage $N - 1$ by a fixed amount. Every posterior variance tends to zero. A Gaussian comparison then makes the selection probability of stage $N - 1$ tend to zero, and the bounded martingale bridge gives the same limit for its empirical frequency.

Suppose stages above k have vanished for both players. Their empirical frequencies vanish as well, leaving the global mixed contrast with limiting lower bound δ_k . Condition (D.5) transfers that advantage to the posterior mixed contrast, and another Gaussian comparison eliminates stage k . Descending through the remaining stages proves (D.6). Once the opponent's selection probability tends to stage 0, every infinite action clock in (D.3) observes stage 0 asymptotically. Equation (D.2) then gives (D.7). $\square$

What remains unproved is the occurrence of $\mathcal{D}_{\text{mix}}^{\text{TS}}$. Infinite sampling and vanishing posterior variance do not settle it because posterior means still average rewards over different action clocks. Theorem D.1 is therefore conditional.

D.3. Regret on the pure-limit branch

Let $\mathcal{C}_{\text{pure}}$ be the event that at least one player's selection distribution converges to a point mass. On this branch, the action-clock law makes the other player converge to a pure best response.

Against pure stage r , the unique best response is $b(r) = \max\{r - 1, 0\}$. Applying the same argument back to the first player requires $r = b(b(r))$, whose only solution is $r = 0$. Thus both players converge to stage 0 on $\mathcal{C}_{\text{pure}}$.

For a Gaussian Thompson arm with limiting payoff gap $\Delta_a > 0$, the exact count law is

$$\frac{N_{i,a}(T)}{\log T} \longrightarrow \frac{2\sigma^2}{\Delta_a^2} \quad \text{almost surely.} \quad (\text{D.8})$$

The upper bound follows from a Gaussian tail bound once the posterior means are separated. A matching lower bound follows from a localized Gaussian lower-tail hazard: if the arm's count were too small, its remaining variance would make another selection sufficiently likely. Stopping the argument before the required posterior inequalities fail avoids conditioning current choices on a future convergence event.

On $\mathcal{C}_{\text{pure}}$, every positive stage has limiting gap one. Simultaneous nonzero play occurs only finitely often, and stage 0 is eventually the best fixed stage in hindsight.

Theorem D.2 (Thompson pure-branch pathwise regret). *On one common probability-one set, $\mathcal{C}_{\text{pure}}$ implies, for both players,*

$$\frac{R_i^{\text{ext}}(T)}{\log T} \longrightarrow 2(N - 1)\sigma^2. \quad (\text{D.9})$$

Proof. Equation (D.8) gives $2\sigma^2 \log T + o(\log T)$ selections of each of the $N - 1$ positive stages. Against opponent stage 0, each such selection loses exactly one raw payoff unit. The finite simultaneous-nonzero contribution is negligible, so summing the count laws gives (D.9).

□

Whether $\mathcal{C}_{\text{pure}}$ occurs remains open. Theorem D.2 states what follows on that branch; it gives neither an unconditional nor an expected logarithmic-regret theorem.

D.4. The two-stage expected-regret countertheorem

Use the literal two-stage configuration from the computational model: prior mean 0, prior variance 100, and pseudo-variance $\sigma^2 = 5/2$. The payoff matrix is

$$M_2 = \begin{pmatrix} 2 & 5 \\ 1 & 4 \end{pmatrix}. \quad (\text{D.10})$$

Stage 0 beats stage 1 by one against either opponent stage, so raw external regret again equals the player's stage-one count.

Theorem D.3 (Thompson expected-regret countertheorem). *For every supplied exact same-*

policy Gaussian Thompson process with the parameters above, there is a deterministic sequence $T_n \rightarrow \infty$ and a process-dependent constant $c > 0$ such that, for both players and every $n \geq 10000$,

$$\mathbb{E}[R_i^{\text{ext}}(T_n)] \geq cT_n^{1/365}. \quad (\text{D.11})$$

Consequently, no constants C and T_0 satisfy $\mathbb{E}[R_i^{\text{ext}}(T)] \leq C \log T$ for every $T \geq T_0$.

Proof. I establish the lower bound with a finite-prefix metastable event. Let the first 100 joint actions be $(1, 0)$, the next n be $(0, 0)$, the next $100n$ be $(1, 1)$, and the following

$$H_n = \left\lfloor \exp\left(\frac{73}{100}n\right) \right\rfloor \quad (\text{D.12})$$

also be $(1, 1)$. Put

$$T_n = 100 + 101n + H_n. \quad (\text{D.13})$$

The first block has a fixed positive Gaussian-cylinder probability. During the $(0, 0)$ block, the exact posterior states imply that one player selects stage 0 with probability greater than $31/32$ and the other with probability at least $1/2$ after the fixed initial prefix. The block therefore has probability at least a fixed positive constant times $(31/64)^n$.

The first 1000 rounds of the $(1, 1)$ block form another fixed positive cylinder. Gaussian Chernoff bounds control the remaining hazards: the middle part survives with factor at least $497/500$, the late part with factor at least $97/100$, and a sequential union bound gives probability at least $1/2$ of remaining at $(1, 1)$ through the final H_n rounds. All conditioning is on finite-history prefix events before the next action. The complete event E_n therefore satisfies

$$\Pr(E_n) \geq c_0 \left(\frac{31}{64}\right)^n \quad (\text{D.14})$$

for a constant $c_0 > 0$. On E_n , both players select stage 1 throughout the final block, so each incurs at least H_n external regret. It follows that

$$\mathbb{E}[R_i^{\text{ext}}(T_n)] \geq c_0 H_n \left(\frac{31}{64}\right)^n. \quad (\text{D.15})$$

The exact inequality

$$\frac{31}{64} \exp\left[\frac{73}{100}\left(1 - \frac{1}{365}\right)\right] > 1 \quad (\text{D.16})$$

and the bounds $H_n \geq \frac{1}{2} \exp(73n/100)$ and $T_n \leq 2 \exp(73n/100)$ for $n \geq 10000$ convert

(D.15) into (D.11). The ratio $T^{1/365}/\log T$ diverges, so an eventual all-horizon logarithmic upper bound would fail on this subsequence. $\square$

Theorem D.3 refutes only the old unconditional expected $O(\log T)$ claim for the notebook's two-stage configuration. It leaves nonconvergence, linear expected regret, and failure of expected no-regret unproved.

Appendix E: Exact Publication Boundary

The three policies do not support one theorem.

1. Repaired anytime Exp3 converges unconditionally in supplied same-policy self-play for $0 < b \leq 1/2$. For $0 < b < 1/2$, raw external regret divided by the exploration clock tends almost surely to $(N - 1)/N$. The expected limit requires equal unit initialization. At $b = 1/3$, both limits have coefficient $3d(N - 1)/(2N)$ on the $T^{2/3}$ scale.
2. Zero-initialized sample-mean epsilon-greedy has almost-sure infinite sampling and action-clock regularity. The mixed calibration event implies stage-zero selection and empirical convergence, but its occurrence is not proved. Permanent greedy-set settlement implies the pathwise logarithmic regret coefficient $(N - 1)/N$. In the exact two-stage process, expected external regret is at least $\sqrt{T}/2$, so no unconditional expected logarithmic upper bound exists.
3. Gaussian Thompson Sampling has almost-sure infinite sampling, vanishing posterior variances, and action-clock regularity. Its mixed calibration event implies stage-zero selection and empirical convergence, but its occurrence is not proved. On the pure-selection-limit branch, the pathwise logarithmic regret coefficient is $2(N - 1)\sigma^2$. In the notebook's two-stage configuration, expected external regret exceeds $cT_n^{1/365}$ along a deterministic subsequence, ruling out an eventual logarithmic upper bound.

These statements concern current selection probabilities, empirical play, and raw external regret against a fixed stage on the realized opponent sequence. They do not establish mixed-policy convergence, eventual realized-action constancy, policy regret, or equilibrium payoff shortfall.